

**DATA-DRIVEN ANALYSIS OF FISCAL STRESS IN KERALA:
REVENUE TRENDS, ECONOMIC GROWTH, AND FISCAL
SUSTAINABILITY (2005–2025)**

Undertaken At

Department Of Economics & Statistics, Malappuram

Guided By

SEBIN MATHEW

Research Officer

Submitted By

FATHIMA LINA

St. Teresa's College (Autonomous), Ernakulam

Affiliated to M G University

DECLARATION

I hereby declare that the project report titled “Data-Driven Analysis of Fiscal Stress in Kerala: Revenue Trends, Economic Growth, And Fiscal Sustainability (2005–2025)” is the outcome of my original work carried out during the Internship at the Department of Economics and Statistics, Malappuram.

I further declare that this report has not been submitted, either in part or full, for the award of any other degree or diploma in any university or institution.

All information and data presented in this report have been collected from authentic and official sources, and due acknowledgement has been made whenever necessary.

Place: Malappuram

Date:

Name: FATHIMA LINA

Signature:

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to all those who supported me throughout the completion of this project.

First and foremost, I extend my heartfelt thanks to the Department of Economics and Statistics, Malappuram, for providing me with the opportunity to undertake this internship and access to valuable data for my study.

I am deeply grateful to my supervisor, Sebin Mathew, for their constant support, insightful suggestions, and guidance throughout the project.

ABSTRACT

Fiscal sustainability is essential for maintaining stable public finances and supporting long-term economic development. Kerala, despite its strong social development achievements and extensive public welfare commitments, has experienced increasing fiscal pressures arising from rising expenditure, widening fiscal deficits, and changing economic conditions over the past two decades. Assessing these fiscal dynamics is therefore important for understanding the state's financial health and sustainability.

The present study, examines Kerala's fiscal performance over the period 2005–2025 using secondary data obtained from the *Kerala Economic Review*. The analysis considers major fiscal indicators including Own Tax Revenue, Own Non-Tax Revenue, Revenue Receipt, Revenue Expenditure, Capital Expenditure, Total Expenditure, Revenue Deficit, Fiscal Deficit and Gross State Domestic Product (GSDP). Fiscal ratios such as Revenue Receipt Ratio, Revenue Expenditure Ratio, Capital Expenditure Ratio, Total Expenditure Ratio, Revenue Deficit Ratio, Fiscal Deficit Ratio and Capital Expenditure as a Percentage of Total Expenditure are also analysed to evaluate fiscal sustainability and expenditure composition.

The study employs descriptive statistics, trend analysis, Compound Annual Growth Rate (CAGR), growth analysis, ratio analysis, correlation analysis, period-wise analysis and a Fiscal Stress Score based on standardized fiscal indicators. The period-wise analysis compares fiscal performance across the Pre-GST, GST, COVID and Recovery phases to assess the impact of major policy changes and economic disruptions.

The findings indicate a substantial increase in revenue mobilisation, government expenditure and GSDP during the study period. While Own Tax Revenue and Revenue Receipts exhibited consistent long-term growth, expenditure expanded at a relatively faster pace, resulting in persistent revenue and fiscal deficits. The ratio analysis suggests that although Kerala maintained relatively stable expenditure patterns, fiscal stress intensified during the COVID period due to increased expenditure commitments and weakened fiscal balances. The Fiscal Stress Score further identifies years with comparatively higher and lower fiscal stress, highlighting the influence of economic shocks and fiscal policy changes on the state's financial position.

Overall, the study provides a comprehensive data-driven assessment of Kerala's fiscal performance and offers useful insights into the state's fiscal sustainability, expenditure management and evolving fiscal challenges over the last two decades.

Keywords: Fiscal Stress, Fiscal Sustainability, Revenue Receipt, Revenue Expenditure, Fiscal Deficit, Revenue Deficit, Capital Expenditure, Total Expenditure, GSDP, Fiscal Ratios, Kerala.

CONTENTS

CHAPTER 1	1
INTRODUCTION.....	1
1.1 Background of the Study	1
1.2 Significance of the Study	2
1.3 Objectives of the Study	2
1.4 Scope of the Study	3
1.5 Limitations of the Study	4
1.6 Chapter Scheme	4
CHAPTER 2	5
DATA DESCRIPTION AND METHODOLOGY.....	5
2.1 Data Description.....	5
2.2 Period Classification	5
2.3 Methodology	6
2.3.1 Descriptive Statistics	6
2.3.2 Trend Analysis	6
2.3.3 Compound Annual Growth Rate (CAGR)	6
2.3.4 Growth Analysis	7
2.3.5 Ratio Analysis	7
2.3.6 Correlation Analysis	8
2.3.7 Fiscal Stress Score	8
CHAPTER 3	10
ANALYSIS AND RESULTS	10
3.1 Descriptive Statistics Analysis.....	10
3.2 Trend Analysis	11
3.3 Correlation Analysis	22
3.4 Period-wise Analysis	23
3.5 Growth Analysis	25
3.6 Fiscal Stress Score Analysis.....	27
CHAPTER 4.....	31

FINDINGS AND CONCLUSIONS	31
4.1 Major Findings.....	31
4.2 Conclusion	32
4.3 Suggestions.....	33
REFERENCES.....	35

CHAPTER 1

INTRODUCTION

1.1 Background of the Study

Public finance is one of the fundamental pillars of economic development, as it enables governments to mobilize resources, provide essential public services, invest in infrastructure, and promote inclusive growth. A sound fiscal system ensures that revenue generation keeps pace with expenditure requirements while maintaining long-term fiscal sustainability. In recent years, managing public finances has become increasingly challenging for state governments due to rising expenditure commitments, changing economic conditions, policy reforms, and unforeseen economic shocks such as the COVID-19 pandemic.

Kerala has long been recognised for its remarkable achievements in education, healthcare, social welfare, and human development. Sustaining these achievements requires substantial public investment and efficient fiscal management. Over the years, the state has experienced continuous growth in revenue mobilisation as well as public expenditure. However, increasing expenditure commitments, persistent revenue and fiscal deficits, and growing pressure on government finances have raised concerns regarding the long-term sustainability of Kerala's fiscal position. Consequently, understanding the state's fiscal performance has become increasingly important for evaluating its financial stability and development prospects.

The structure of Kerala's economy also plays an important role in shaping its fiscal performance. The state's Gross State Value Added (GSVA) is predominantly driven by the tertiary sector, which accounted for more than two-thirds of the total GSVA in recent years, while the secondary sector contributed nearly one-fourth and the primary sector accounted for less than one-tenth of the economy. This sectoral composition reflects Kerala's service-oriented economy, where activities such as trade, tourism, transport, financial services, real estate, education, and healthcare contribute significantly to economic growth. Since government revenue largely depends on the level and composition of economic activity, changes in sectoral performance directly influence the state's fiscal capacity.

Fiscal stress arises when the growth in government expenditure consistently outpaces the growth in available revenue resources, thereby reducing the government's ability to finance developmental activities without increasing fiscal imbalances. Persistent fiscal stress may result in higher revenue and fiscal deficits, reduced fiscal flexibility, and greater dependence on borrowing. Assessing fiscal stress therefore provides valuable insights into the financial health of a government and its ability to maintain fiscal sustainability over time.

Against this background, the present study, titled "**Data-Driven Analysis of Fiscal Stress in Kerala: Revenue Trends, Economic Growth, and Fiscal Sustainability (2005–2025)**," examines Kerala's fiscal performance over the period 2005–2025. The study analyses major

fiscal indicators including Own Tax Revenue, Own Non-Tax Revenue, Revenue Receipt, Revenue Expenditure, Capital Expenditure, Total Expenditure, Revenue Deficit, Fiscal Deficit, and Gross State Domestic Product (GSDP). It further evaluates fiscal sustainability using key fiscal ratios, namely Revenue Receipt Ratio, Revenue Expenditure Ratio, Capital Expenditure Ratio, Total Expenditure Ratio, Revenue Deficit Ratio, Fiscal Deficit Ratio, and Capital Expenditure as a Percentage of Total Expenditure. Through descriptive statistics, trend analysis, growth analysis, period-wise comparison, correlation analysis, and fiscal stress assessment, the study seeks to provide a comprehensive understanding of Kerala's evolving fiscal condition and the sustainability of its public finances over the last two decades.

1.2 Significance of the Study

Fiscal sustainability has become an important concern for both policymakers and researchers as state governments face increasing pressure to balance revenue generation with growing expenditure commitments. In Kerala, maintaining fiscal discipline while ensuring continued investment in public welfare and infrastructure has become particularly challenging due to changing economic conditions, policy reforms, and external shocks such as the COVID-19 pandemic. Evaluating fiscal performance is therefore essential for understanding the state's financial stability and long-term development prospects.

The present study is significant because it provides a comprehensive, data-driven assessment of Kerala's fiscal performance over the period 2005–2025. Unlike studies that focus only on revenue or expenditure, this study examines multiple dimensions of public finance by analysing revenue mobilisation, expenditure patterns, fiscal deficits, economic growth, expenditure composition, and fiscal sustainability using a broad set of fiscal indicators and ratios.

The study also compares Kerala's fiscal performance across four distinct phases—Pre-GST, GST, COVID, and Recovery—to understand how major policy changes and economic disruptions influenced the state's fiscal position. In addition, the Fiscal Stress Score provides a composite measure for identifying periods of relatively higher and lower fiscal stress, offering a clearer understanding of the evolution of Kerala's public finances.

The findings of the study are expected to be useful for policymakers, researchers, academicians, students, and public administrators by providing empirical evidence on Kerala's fiscal trends and sustainability. The study may also serve as a reference for future research on state finances and fiscal management in India.

1.3 Objectives of the Study

General Objective

To analyse fiscal stress and fiscal sustainability in Kerala by examining revenue trends, expenditure patterns, economic growth, and key fiscal indicators during the period 2005–2025.

Specific Objectives

1. To examine the trends and growth patterns of major fiscal indicators, including Own Tax Revenue, Own Non-Tax Revenue, Revenue Receipt, Revenue Expenditure, Capital Expenditure, Total Expenditure, Revenue Deficit, Fiscal Deficit, and Gross State Domestic Product (GSDP) during 2005–2025.
2. To evaluate Kerala's fiscal sustainability using Revenue Receipt Ratio, Revenue Expenditure Ratio, Capital Expenditure Ratio, Total Expenditure Ratio, Revenue Deficit Ratio, Fiscal Deficit Ratio, and Capital Expenditure as a Percentage of Total Expenditure.
3. To examine the relationship between major fiscal indicators and economic growth using correlation analysis.
4. To compare Kerala's fiscal performance across the Pre-GST, GST, COVID, and Recovery periods through period-wise analysis.
5. To assess fiscal stress by constructing a Fiscal Stress Score and identifying periods of relatively higher and lower fiscal stress during the study period.

1.4 Scope of the Study

The study examines the fiscal performance of the Government of Kerala over the period **2005–2025** using secondary data obtained from the *Kerala Economic Review*. It focuses on analysing the state's revenue mobilisation, expenditure pattern, fiscal deficits, and economic growth to understand the changing nature of fiscal stress and fiscal sustainability over the last two decades.

The analysis covers major fiscal indicators, namely Own Tax Revenue, Own Non-Tax Revenue, Revenue Receipt, Revenue Expenditure, Capital Expenditure, Total Expenditure, Revenue Deficit, Fiscal Deficit, and Gross State Domestic Product (GSDP). In addition, the study evaluates fiscal sustainability using Revenue Receipt Ratio, Revenue Expenditure Ratio, Capital Expenditure Ratio, Total Expenditure Ratio, Revenue Deficit Ratio, Fiscal Deficit Ratio, and Capital Expenditure as a Percentage of Total Expenditure.

To achieve the objectives of the study, statistical techniques such as descriptive statistics, trend analysis, Compound Annual Growth Rate (CAGR), growth analysis, ratio analysis, correlation analysis, period-wise analysis, and Fiscal Stress Score are employed. The study also compares Kerala's fiscal performance across the Pre-GST, GST, COVID, and Recovery periods to assess the impact of major policy changes and economic disruptions on the state's fiscal condition.

The study is confined to the selected fiscal indicators and the specified study period. It does not include an analysis of public debt, contingent liabilities, off-budget borrowings, or other macroeconomic factors beyond the scope of the selected variables.

1.5 Limitations of the Study

The study has the following limitations:

1. The study is based entirely on secondary data collected from the *Kerala Economic Review*.
2. The analysis is confined to the fiscal performance of Kerala and covers the period from **2005 to 2025**.
3. The assessment of fiscal stress is based on selected fiscal indicators and ratios; therefore, it may not capture all dimensions of the State's fiscal condition.
4. The study does not include an analysis of public debt, contingent liabilities, off-budget borrowings, or other macroeconomic factors beyond the selected variables.
5. The findings and conclusions are limited to the scope of the selected data and methodology adopted in the study.

1.6 Chapter Scheme

The study is organised into four chapters.

Chapter 1 introduces the study by presenting the background, significance, objectives, scope, limitations, and chapter scheme. Chapter 2 describes the data used for the study, the classification of the study period, and the statistical methods employed, including descriptive statistics, trend analysis, Compound Annual Growth Rate (CAGR), growth analysis, ratio analysis, correlation analysis, multiple regression analysis, and Fiscal Stress Score. Chapter 3 presents the analysis and interpretation of the data. It includes descriptive statistics, trend analysis, correlation analysis, period-wise analysis, regression analysis, growth analysis, fiscal stress assessment, and the interpretation of the major findings based on the selected fiscal indicators. Chapter 4 summarises the major findings of the study, presents the conclusions drawn from the analysis, and offers suggestions for improving fiscal management and sustainability in Kerala.

CHAPTER 2

DATA DESCRIPTION AND METHODOLOGY

2.1 Data Description

The study is based entirely on secondary data covering the period from **2005–06 to 2024–25**. The data were collected from various editions of the *Kerala Economic Review*, published by the State Planning Board, Government of Kerala. The Economic Review serves as one of the principal sources of information on Kerala's economy and provides comprehensive statistics on the state's public finance, economic growth, and development.

The dataset comprises annual observations on major fiscal and economic indicators of Kerala. The variables included in the study are **Own Tax Revenue (OTR)**, **Own Non-Tax Revenue (ONTR)**, **Revenue Receipt (RR)**, **Revenue Expenditure (RE)**, **Capital Expenditure (CE)**, **Total Expenditure (TE)**, **Revenue Deficit (RD)**, **Fiscal Deficit (FD)**, and **Gross State Domestic Product (GSDP) at Current Prices**.

To assess fiscal sustainability, the study further constructs a set of fiscal ratios by expressing selected fiscal indicators as percentages of GSDP. These include the Revenue Receipt Ratio (RR Ratio), Revenue Expenditure Ratio (RE Ratio), Capital Expenditure Ratio (CE Ratio), Total Expenditure Ratio (TE Ratio), Revenue Deficit Ratio (RD Ratio), and Fiscal Deficit Ratio (FD Ratio). In addition, Capital Expenditure as a Percentage of Total Expenditure is calculated to examine the share of developmental expenditure within the state's overall expenditure.

The compiled dataset forms the basis for the statistical analyses carried out in the study, including descriptive statistics, trend analysis, Compound Annual Growth Rate (CAGR), growth analysis, ratio analysis, correlation analysis, period-wise analysis, and fiscal stress assessment. The selected variables provide a comprehensive framework for evaluating Kerala's revenue mobilisation, expenditure pattern, fiscal balance, economic growth, and overall fiscal sustainability during the study period.

2.2 Period Classification

For comparative analysis, the study period was divided into four distinct phases based on major policy and economic developments.

Period	Years
Pre-GST Period	2005–06 to 2016–17
GST Period	2017–18 to 2019–20

Period	Years
COVID-19 Period	2020–21 to 2021–22
Recovery Period	2022–23 to 2024–25

This classification helps in examining the impact of tax reforms, economic disruptions, and recovery measures on Kerala's fiscal performance.

2.3 Methodology

To evaluate fiscal stress and fiscal performance in Kerala, various statistical and analytical techniques were employed.

2.3.1 Descriptive Statistics

Descriptive statistics such as Mean, Median, Standard Deviation, Minimum, and Maximum values were calculated for all major variables. These measures provide an overview of the central tendency and variability of fiscal indicators during the study period.

2.3.2 Trend Analysis

Trend analysis was carried out using line graphs to examine the movement of major fiscal and economic indicators over the study period. It was used to identify the long-term pattern of variables such as Own Tax Revenue, Revenue Receipt, Revenue Expenditure, Capital Expenditure, Total Expenditure, Revenue Deficit, Fiscal Deficit, Gross State Domestic Product (GSDP), Capital Expenditure % and the selected fiscal ratios. The analysis helps in understanding the overall direction and behaviour of Kerala's fiscal performance from **2005–06 to 2024–25**.

2.3.3 Compound Annual Growth Rate (CAGR)

The Compound Annual Growth Rate (CAGR) was used to measure the average annual growth of fiscal variables over the study period.

$$\text{CAGR} = [(\text{Ending Value} / \text{Beginning Value})^{(1/n)}] - 1$$

where:

- Ending Value = Value in the final year
- Beginning Value = Value in the initial year
- n = Number of years

CAGR provides a standardized measure of long-term growth by accounting for the compounding effect over time.

2.3.4 Growth Analysis

Growth analysis was conducted to measure the year-on-year percentage change in major fiscal and economic indicators. The annual growth rates of Own Tax Revenue, Revenue Receipt, Revenue Expenditure, Capital Expenditure, Fiscal Deficit, and Gross State Domestic Product (GSDP) were calculated to examine changes in fiscal performance over time. This analysis helps identify periods of rapid growth, slowdown, or decline and provides insights into the fluctuations in Kerala's fiscal indicators during the study period.

2.3.5 Ratio Analysis

To assess fiscal performance relative to the size of the economy, the following fiscal ratios were computed:

Revenue Receipt Ratio

$$\text{Revenue Receipt Ratio} = (\text{Revenue Receipt} / \text{GSDP}) \times 100$$

Revenue Expenditure Ratio

$$\text{Revenue Expenditure Ratio} = (\text{Revenue Expenditure} / \text{GSDP}) \times 100$$

Capital Expenditure Ratio

$$\text{Capital Expenditure Ratio} = (\text{Capital Expenditure} / \text{GSDP}) \times 100$$

Total Expenditure Ratio

$$\text{Total Expenditure Ratio} = (\text{Total Expenditure} / \text{GSDP}) \times 100$$

Revenue Deficit Ratio

$$\text{Revenue Deficit Ratio} = (\text{Revenue Deficit} / \text{GSDP}) \times 100$$

Fiscal Deficit Ratio

$$\text{Fiscal Deficit Ratio} = (\text{Fiscal Deficit} / \text{GSDP}) \times 100$$

Capital Expenditure as a Percentage of Total Expenditure

$$\text{Capital Expenditure as \% of Total Expenditure} = (\text{Capital Expenditure} / \text{Total Expenditure}) \times 100$$

These ratios measure the relative size of government revenue, expenditure, and fiscal deficits with respect to the state's economy. They provide a standardized basis for comparing Kerala's fiscal performance across different years and help assess the sustainability and expenditure composition of the state's public finances.

2.3.6 Correlation Analysis

Pearson's correlation coefficient was used to examine the relationship among fiscal variables and economic growth indicators. Correlation analysis helps determine the strength and direction of association between revenue, expenditure, deficit, and GSDP. The results provide insights into how changes in one fiscal variable are related to changes in another.

2.3.7 Fiscal Stress Score

The Fiscal Stress Score was developed as a composite indicator to assess the overall fiscal condition of Kerala during the study period. The score was constructed using six fiscal ratios: **Revenue Receipt Ratio (RR Ratio)**, **Revenue Expenditure Ratio (RE Ratio)**, **Revenue Deficit Ratio (RD Ratio)**, **Capital Expenditure Ratio (CE Ratio)**, **Total Expenditure Ratio (TE Ratio)**, and **Fiscal Deficit Ratio (FD Ratio)**.

Since these ratios are measured on different scales, they were first standardized using the **Z-score transformation** to make them comparable. The standardized value for each ratio was calculated as:

$$Z = \frac{X - \bar{X}}{SD}$$

where,

- **Z** = Standardized (Z-score) value
- **X** = Actual value of the ratio
- **$\bar{X}$** = Mean of the ratio
- **SD** = Standard deviation of the ratio

The Fiscal Stress Score for each year was obtained by taking the arithmetic mean of the six standardized ratios:

$$\text{Fiscal Stress Score} = \frac{Z_{RR} + Z_{RE} + Z_{RD} + Z_{CE} + Z_{TE} + Z_{FD}}{6}$$

The calculated Fiscal Stress Scores were then ranked to identify the years with relatively **higher** and **lower** levels of fiscal stress. This composite measure provides an overall

assessment of Kerala's fiscal position by combining multiple dimensions of revenue generation, expenditure, and fiscal deficits into a single indicator.

CHAPTER 3

ANALYSIS AND RESULTS

3.1 Descriptive Statistics Analysis

Descriptive statistics provide a summary of the major characteristics of the selected fiscal variables over the study period. Measures such as the mean, median, standard deviation, minimum and maximum values help understand the central tendency, variability and overall distribution of the data.

Table 3.1.1 Descriptive Statistics of Major Fiscal Indicators in Kerala (2005–06 to 2024–25)

VARIABLE	MEAN	MEDIAN	STANDARD DEVIATION	MINIMUM	MAXIMUM	CAGR
OWN TAX REVENUE	38564.46	37113.83	20974.40	9779.64	76642.20	11.44%
OWN NON-TAX REVENUE	7359.48	7305.50	5373.17	936.77	16486.62	
REVENUE RECEIPT	66627.41	63491.57	39478.70	15294.52	132724.65	11.68%
REVENUE EXPENDITURE	79241.02	75217.95	46787.22	18423.58	155920.79	11.90%
CAPITAL EXPENDITURE	8459.33	7050.40	5765.63	1104.06	17886.77	15.79%
TOTAL EXPENDITURE	87700.36	81887.94	52451.43	19527.64	173807.56	12.19%
REVENUE DEFICIT	12613.62	10482.69	8660.17	2637.95	31059.72	
FISCAL DEFICIT	20096.30	18230.06	12556.87	3821.87	48248.14	13.74%
GSDP	576140.76	537278.83	344111.92	124389.42	1248533.01	12.91%
CAPITAL EXPENDITURE %	9.09%	9.43%	0.016	5.65%	11.12%	

Table 3.1.1 presents the descriptive statistics of the selected fiscal indicators of Kerala for the period 2005–06 to 2024–25. Table 3.1 presents the descriptive statistics of the major fiscal and economic indicators of Kerala for the period **2005–06 to 2024–25**. The measures include the **mean, median, standard deviation, minimum, and maximum** values, which provide an overview of the distribution and variability of the selected variables during the study period.

Among the revenue indicators, **Revenue Expenditure** recorded a higher average (₹79,241.02 crore) than **Revenue Receipt** (₹66,627.41 crore), indicating that expenditure consistently exceeded revenue over the study period. **Own Tax Revenue** had a mean value of **₹38,564.46**

crore, whereas **Own Non-Tax Revenue** recorded a comparatively lower mean of **₹7,359.48 crore**, reflecting the greater contribution of tax revenue to the state's own revenue mobilization.

The expenditure indicators exhibit considerable variation over the study period. **Total Expenditure** recorded the highest average value of **₹87,700.36 crore**, followed by **Revenue Expenditure** (₹79,241.02 crore). **Capital Expenditure** had a mean value of **₹8,459.33 crore**, indicating that capital spending constituted a relatively smaller share of the state's total expenditure. On average, **Capital Expenditure accounted for 9.09% of Total Expenditure**, suggesting that a major portion of government expenditure was incurred on revenue-related activities.

Among the deficit indicators, the average **Revenue Deficit** and **Fiscal Deficit** were **₹12,613.62 crore** and **₹20,096.30 crore**, respectively. The higher average Fiscal Deficit reflects the overall borrowing requirement of the government after accounting for both revenue and capital transactions.

Gross State Domestic Product (GSDP) recorded the highest average value (**₹5,76,140.76 crore**) among all the variables, increasing from a minimum of **₹1,24,389.42 crore** to a maximum of **₹12,48,533.01 crore**, indicating substantial economic growth during the study period.

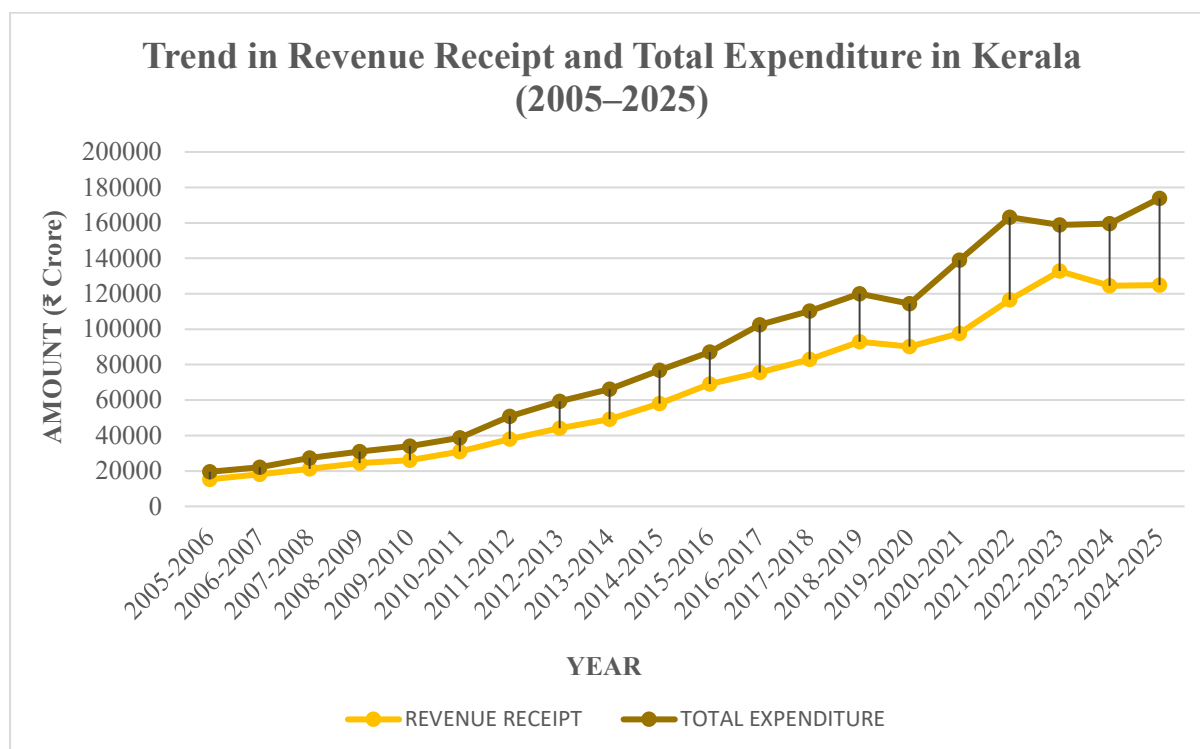
The standard deviation values indicate noticeable variability in the fiscal indicators over time. **Total Expenditure** recorded the highest standard deviation (**52,451.43**), followed by **Revenue Expenditure** (**46,787.22**) and **Revenue Receipt** (**39,478.70**), suggesting considerable fluctuations in government finances across the years. In contrast, **Capital Expenditure as a Percentage of Total Expenditure** exhibited relatively low variation, with values ranging from **5.65%** to **11.12%**, indicating that the share of capital expenditure in total expenditure remained fairly stable despite changes in the absolute level of expenditure.

Overall, the descriptive statistics indicate a steady expansion in Kerala's revenue, expenditure, fiscal deficits, and economic output over the study period. The higher expenditure relative to revenue, together with the persistence of fiscal deficits, highlights the importance of assessing fiscal sustainability through the subsequent analyses presented in this study.

3.2 Trend Analysis

Trend analysis was carried out to examine the movement and long-term behaviour of the selected fiscal indicators over the study period. The graphical presentation of each indicator helps identify patterns of growth, fluctuations and structural changes associated with major economic and policy events. The following figures illustrate the trends in the key fiscal variables of Kerala from 2005–06 to 2024–25.

Figure 3.2.1: Trend in Revenue Receipts and Total Expenditure in Kerala (2005–06 to 2024–25)



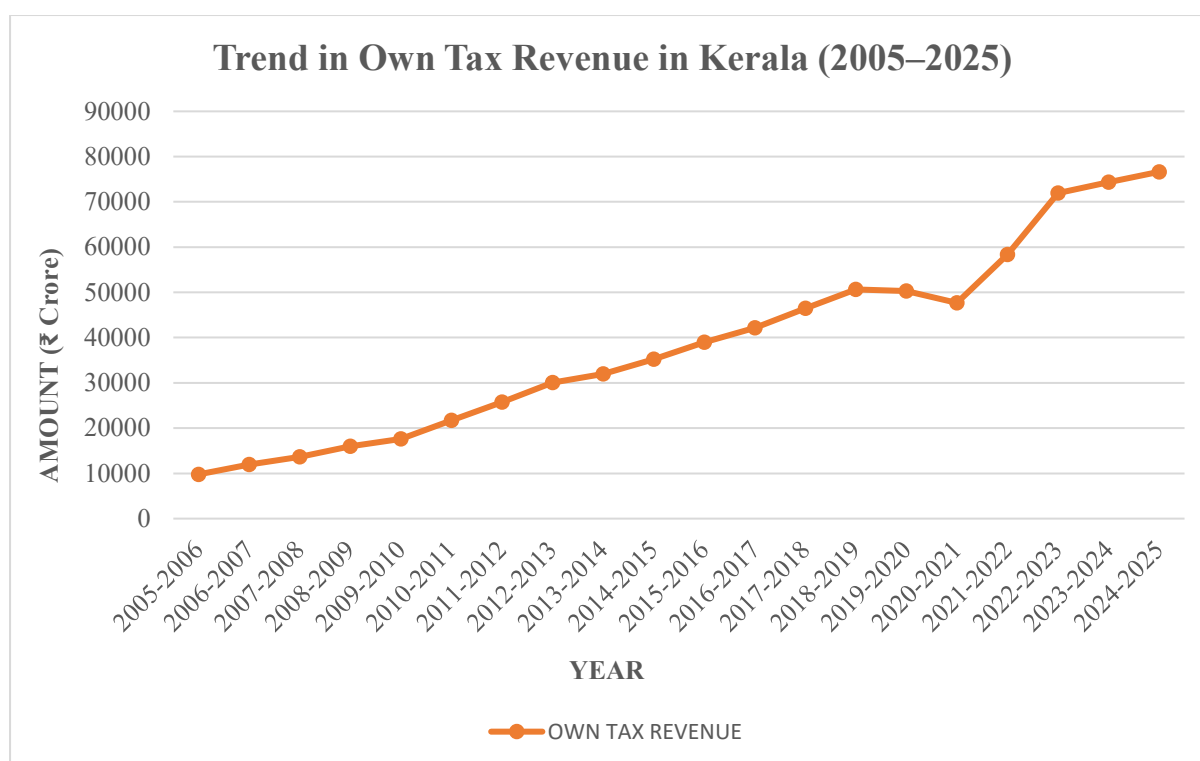
The figure illustrates the trend in **Revenue Receipt** and **Total Expenditure** of Kerala during the period **2005–06 to 2024–25**. Both variables exhibit a generally increasing trend throughout the study period, reflecting the expansion of the state's fiscal operations over time. However, **Total Expenditure remained consistently higher than Revenue Receipt** in every year, indicating that government spending exceeded the revenue generated.

Revenue Receipt increased steadily from **₹15,294.52 crore in 2005–06** to **₹1,24,861.07 crore in 2024–25**, although a slight decline was observed during **2023–24** before stabilising in **2024–25**. The continuous rise in revenue receipts can be attributed to improvements in tax collections, non-tax revenue, central transfers, and grants over the years.

Similarly, **Total Expenditure** showed a sustained upward trend, rising from **₹19,527.64 crore in 2005–06** to **₹1,73,807.56 crore in 2024–25**. The increase became more pronounced after **2020–21**, reflecting higher government spending during the COVID-19 pandemic and the subsequent recovery period. Although expenditure moderated slightly during **2022–23** and **2023–24**, it reached its highest level in **2024–25**.

The widening gap between **Revenue Receipt** and **Total Expenditure** in several years suggests increasing expenditure commitments and continued pressure on the state's finances. The trend indicates that while revenue mobilisation improved considerably during the study period, the growth in expenditure remained relatively higher, highlighting the importance of maintaining fiscal discipline to ensure long-term fiscal sustainability.

Figure 3.2.2: Trend in Own Tax Revenue in Kerala (2005–06 to 2024–25)



The figure presents the trend in **Own Tax Revenue (OTR)** of Kerala during the period **2005–06 to 2024–25**. The overall trend shows a steady increase in Own Tax Revenue over the study period, indicating an improvement in the state's capacity to generate revenue from its own tax sources.

Own Tax Revenue increased from **₹9,779.64 crore** in **2005–06** to **₹76,642.20 crore** in **2024–25**. A consistent upward trend was observed during the Pre-GST period, reflecting growth in the state's tax base and economic activity. Although a slight decline was recorded in **2020–21** due to the disruption caused by the COVID-19 pandemic, Own Tax Revenue recovered strongly in the subsequent years. A sharp increase was observed from **2021–22 onwards**, with revenue crossing **₹70,000 crore** in **2022–23** and continuing to rise during the recovery period.

The sustained growth in Own Tax Revenue indicates an improvement in Kerala's revenue mobilisation efforts and contributed significantly to strengthening the state's revenue base during the later years of the study period.

Figure 3.2.3: Trend in Revenue Deficit in Kerala (2005–06 to 2024–25)



The figure depicts the trend in **Revenue Deficit** in Kerala during the period **2005–06 to 2024–25**. The figure reveals an overall increasing trend with considerable fluctuations, indicating variations in the gap between revenue expenditure and revenue receipts over time.

Revenue Deficit increased from **₹3,129.15 crore** in **2005–06** to **₹31,059.72 crore** in **2024–25**. During the initial years, the increase was gradual, followed by a more pronounced rise after **2011–12**. The deficit remained relatively high during the GST period and continued to increase during the COVID-19 years, reflecting the additional fiscal pressure experienced during the pandemic.

A notable decline in Revenue Deficit was observed in **2023–24**, suggesting a temporary improvement in the state's revenue position. However, the deficit increased sharply again in **2024–25**, reaching the highest level recorded during the study period. The overall pattern indicates that Revenue Expenditure continued to grow faster than Revenue Receipts in several years, highlighting persistent pressure on Kerala's revenue account and the need for sustained fiscal management.

Figure 3.2.4: Trend in Fiscal Deficit in Kerala (2005-2025)

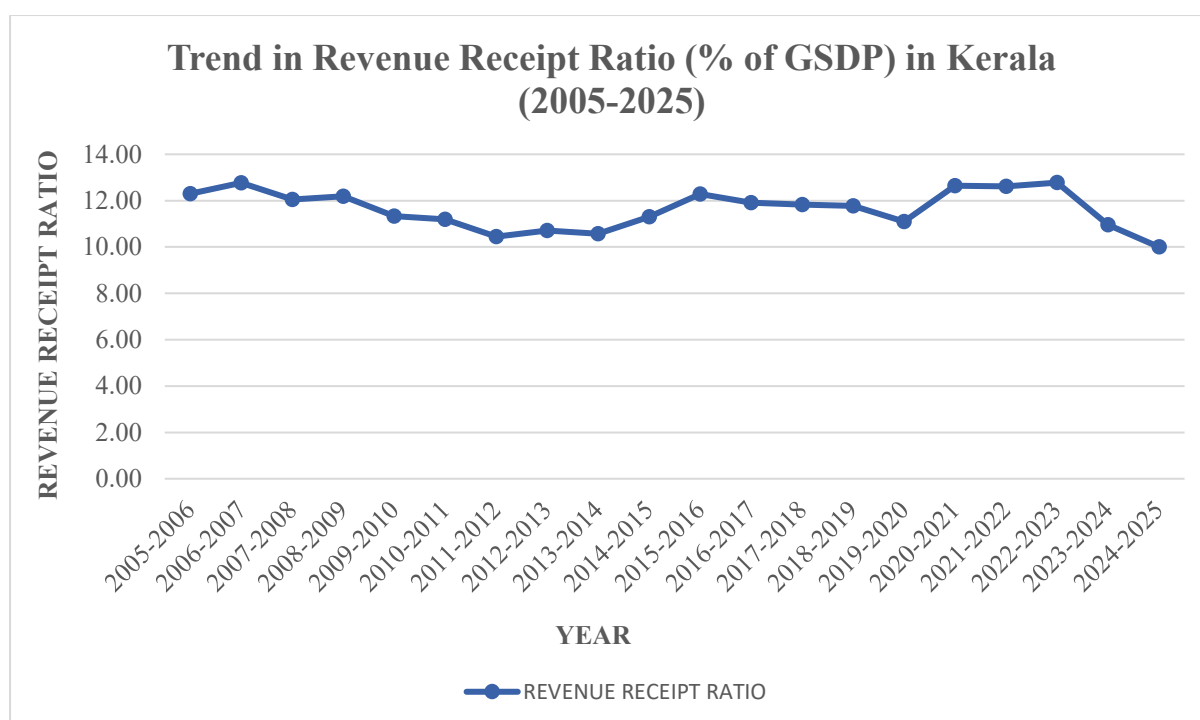


The figure illustrates the trend in **Fiscal Deficit** in Kerala during the period **2005–06 to 2024–25**. The overall trend indicates a substantial increase in the fiscal deficit over the study period, reflecting the growing gap between the government's total expenditure and its total receipts excluding borrowings.

Fiscal Deficit increased from **₹4,181.72 crore** in **2005–06** to **₹48,248.14 crore** in **2024–25**, although the trend was marked by fluctuations across the years. A gradual rise was observed during the initial years, followed by a sharper increase from **2011–12** onwards. The deficit remained relatively high during **2017–18 to 2019–20**, before increasing further during the **COVID-19 period (2020–21 and 2021–22)**, reflecting the combined impact of higher government expenditure and fiscal pressures.

Although Fiscal Deficit declined to **₹25,554.54 crore** in **2022–23**, it increased again during the recovery period, reaching its highest value of **₹48,248.14 crore** in **2024–25**. The persistent increase in Fiscal Deficit indicates rising borrowing requirements to finance government expenditure and highlights the importance of maintaining fiscal discipline while supporting economic growth and public welfare.

Figure 3.2.5: Trend in Revenue Receipt Ratio (% of GSDP) in Kerala (2005-2025)

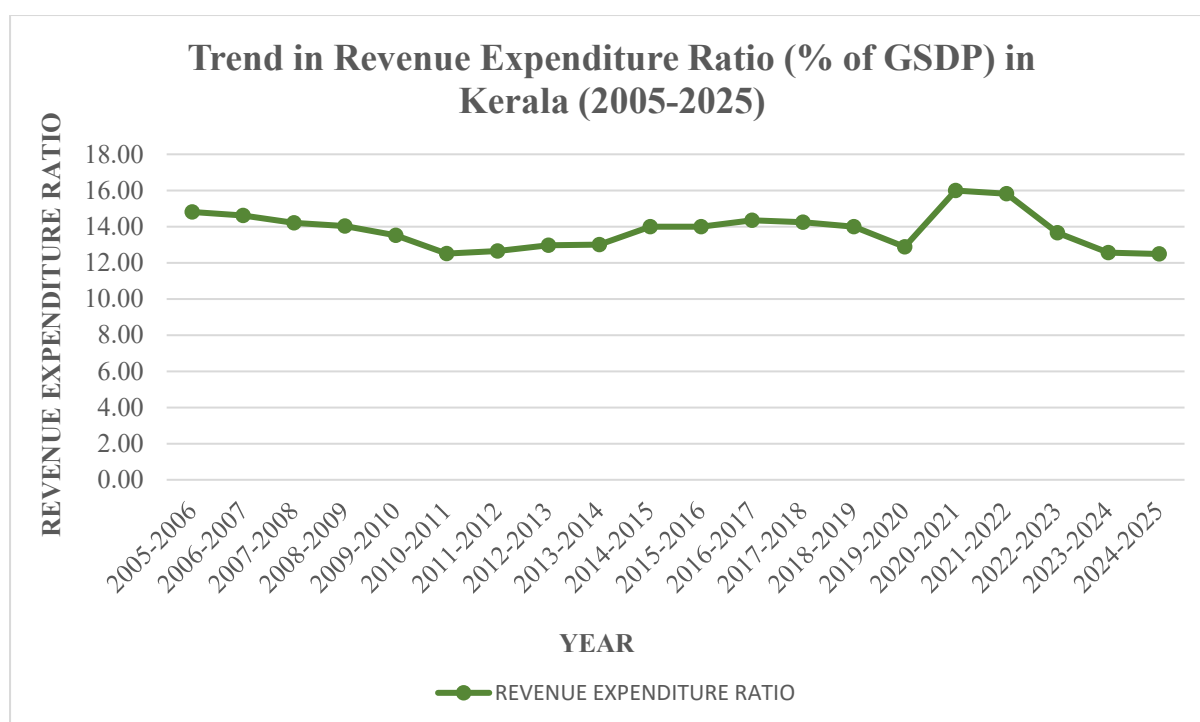


The figure illustrates the trend in **Revenue Receipt Ratio (Revenue Receipts as a percentage of GSDP)** in Kerala during the period **2005–06 to 2024–25**. The ratio indicates the extent to which the state's economy generates revenue receipts for the government relative to its Gross State Domestic Product (GSDP).

Revenue Receipt Ratio showed moderate fluctuations over the study period, remaining within the range of **10.00%** to **12.78%**. The ratio was comparatively higher during the initial years, exceeding **12%** between **2005–06** and **2009–10**. A gradual decline was observed thereafter, reaching around **10.70%** during **2012–13**. From **2015–16 onwards**, the ratio improved and remained relatively stable, before increasing to its highest value of **12.78%** in **2022–23**.

However, the ratio declined during the subsequent years, falling to **10.96%** in **2023–24** and **10.00%** in **2024–25**, the lowest value recorded during the study period. This suggests that although Revenue Receipts continued to increase in absolute terms, the growth of GSDP outpaced the growth in Revenue Receipts during the later years. Overall, Revenue Receipt Ratio remained fairly stable, indicating a consistent capacity of the state to mobilise revenue relative to its economic output.

Figure 3.2.6: Trend in Revenue Expenditure Ratio (% of GSDP) in Kerala (2005-2025)

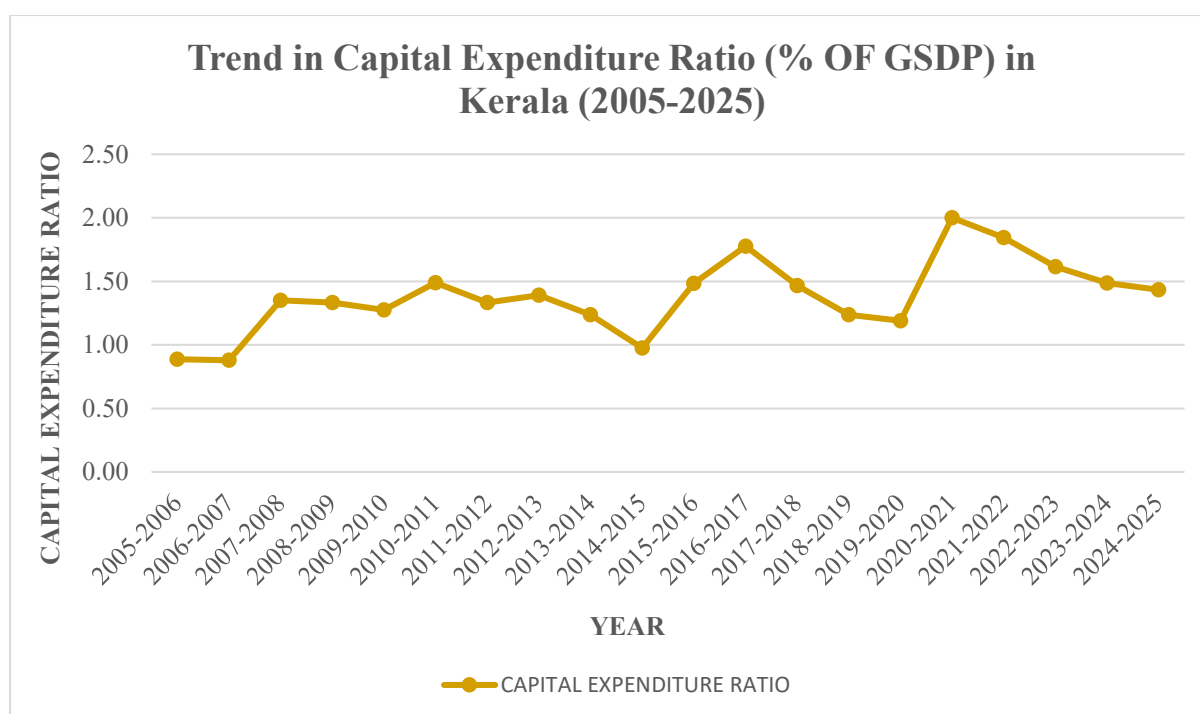


The figure presents the trend in **Revenue Expenditure Ratio (Revenue Expenditure as a percentage of GSDP)** in Kerala during the period **2005–06 to 2024–25**. The ratio measures the share of the state's economic output utilised to meet revenue expenditure.

Revenue Expenditure Ratio generally exhibited a declining trend over the study period, decreasing from **14.81%** in **2005–06** to **12.49%** in **2024–25**. During the Pre-GST period, the ratio gradually declined and remained close to **13%**. A moderate increase was observed between **2015–16** and **2019–20**, indicating relatively higher revenue expenditure in relation to GSDP.

The ratio increased sharply during the **COVID-19 period**, reaching **15.80%** in **2020–21**, the highest value recorded during the study period. This was followed by a gradual decline during the recovery period, falling to **13.67%** in **2022–23**, **12.56%** in **2023–24**, and **12.49%** in **2024–25**. The overall trend suggests that while revenue expenditure remained substantial throughout the study period, its share in GSDP declined over time, reflecting an improvement in expenditure management relative to the growth of the state's economy.

Figure 3.2.7: Trend in Capital Expenditure Ratio (% of GSDP) in Kerala (2005-2025)

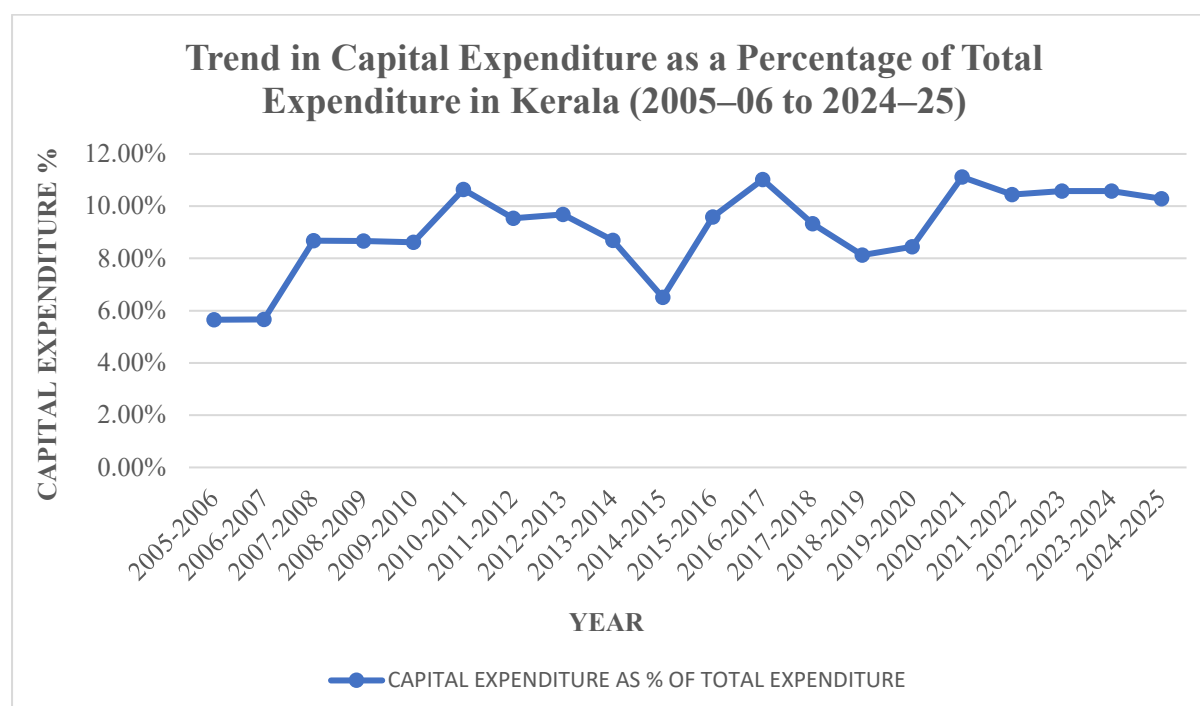


The figure illustrates the trend in **Capital Expenditure Ratio (Capital Expenditure as a percentage of GSDP)** in Kerala during the period **2005–06 to 2024–25**. The ratio indicates the share of the state's economic output allocated towards capital expenditure, which contributes to the creation of long-term public assets and infrastructure.

Capital Expenditure Ratio exhibited noticeable fluctuations throughout the study period, ranging between **0.89%** and **2.00%** of GSDP. The ratio remained below **1%** during the initial years, before increasing gradually to around **1.5%** between **2007–08** and **2013–14**. A decline was observed in **2015–16**, when the ratio fell to **0.98%**, followed by a recovery during the subsequent years.

The ratio reached **2.00%** in **2020–21**, the highest value recorded during the study period, indicating relatively higher capital expenditure in relation to the size of the state's economy during the COVID-19 period. Thereafter, the ratio gradually declined to **1.84%** in **2021–22**, **1.62%** in **2022–23**, **1.49%** in **2023–24**, and **1.43%** in **2024–25**. Overall, the trend suggests that while capital expenditure increased in absolute terms over the years, its share in GSDP fluctuated, reflecting changes in investment priorities and economic growth.

Figure 3.2.8: Trend in Capital Expenditure Ratio (% of Total Expenditure) in Kerala (2005-2025)

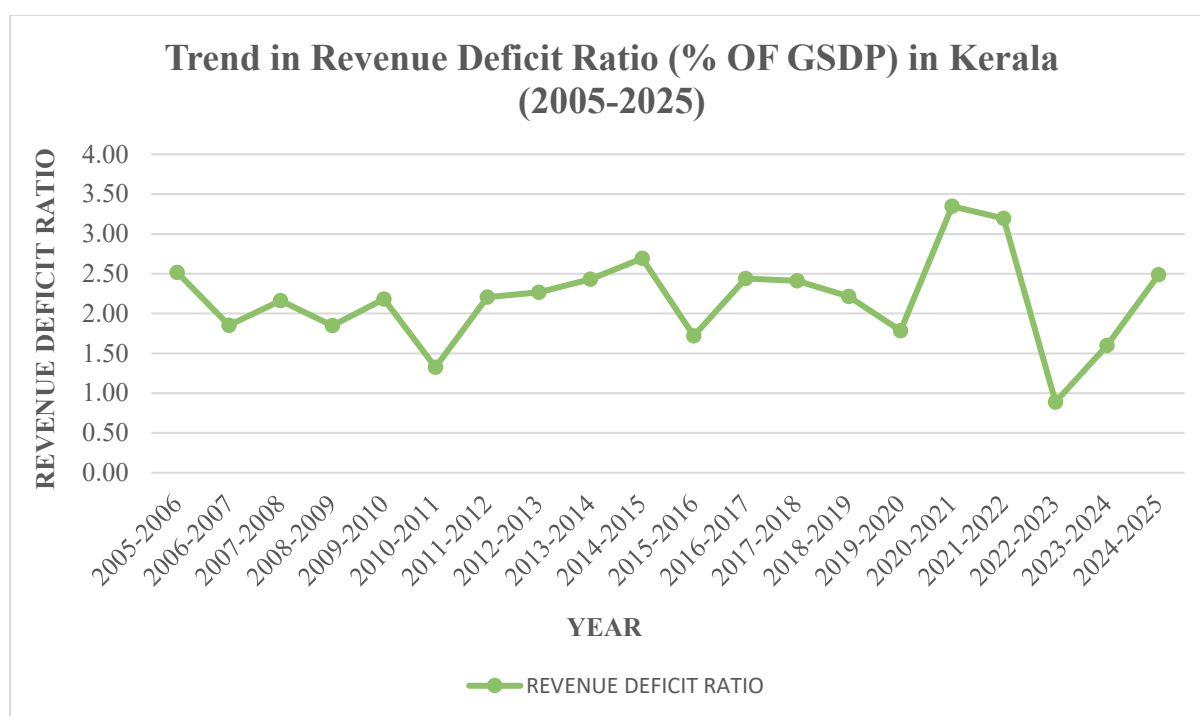


The figure presents the trend in **Capital Expenditure as a Percentage of Total Expenditure** in Kerala during the period **2005–06 to 2024–25**. This ratio indicates the proportion of total government expenditure devoted to capital investment, providing an indication of the state's emphasis on asset creation relative to overall spending.

The share of Capital Expenditure in Total Expenditure showed moderate fluctuations during the study period, varying from **5.65%** to **11.12%**. During **2005–06** and **2006–07**, the ratio remained relatively low at around **5.7%**, after which it increased steadily to around **8–10%** in the following years. A temporary decline was observed in **2014–15**, when the ratio decreased to **6.51%**, before recovering in subsequent years.

The highest share of **11.12%** was recorded in **2020–21**, indicating that a relatively larger proportion of total government expenditure was allocated to capital expenditure during that year. In the years that followed, the ratio remained consistently above **10%**, reaching **10.58%** in both **2022–23** and **2023–24**, before recording **10.29%** in **2024–25**. Overall, the trend indicates a gradual increase in the share of capital expenditure within total expenditure over the study period, reflecting a relatively stronger emphasis on capital investment in the later years.

Figure 3.2.9: Trend in Revenue Deficit Ratio (% of GSDP) in Kerala (2005–2025)

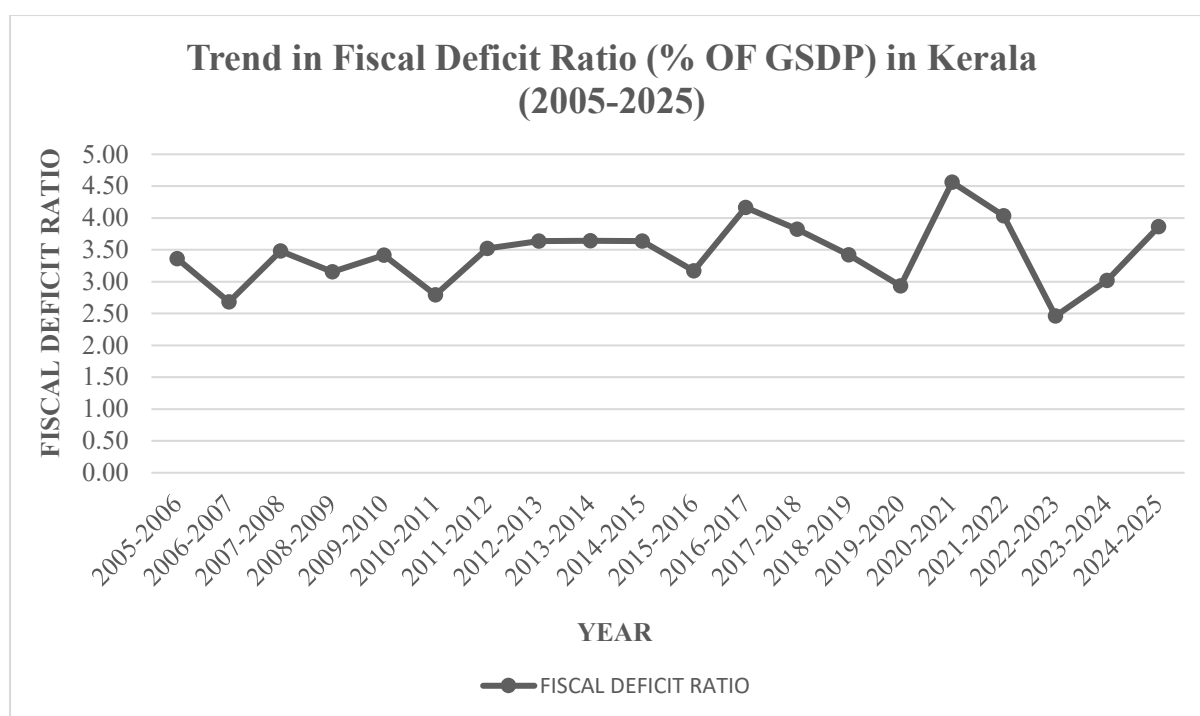


The figure illustrates the trend in **Revenue Deficit Ratio (Revenue Deficit as a percentage of GSDP)** in Kerala during the period **2005–06 to 2024–25**. This ratio reflects the extent to which revenue expenditure exceeds revenue receipts relative to the size of the state's economy.

Revenue Deficit Ratio fluctuated throughout the study period, ranging from **0.89%** to **3.35%**. During the initial years, the ratio remained between **1.8%** and **2.5%**, indicating a moderate level of revenue deficit relative to GSDP. A temporary decline was observed in **2010–11**, when the ratio fell to **1.33%**, followed by a gradual increase over the subsequent years.

The ratio reached its highest level of **3.35%** in **2020–21**, reflecting the severe fiscal impact of the COVID-19 pandemic. Although it declined sharply to **0.89%** in **2022–23**, the ratio increased again to **1.60%** in **2023–24** and **2.49%** in **2024–25**. Overall, the trend indicates that Kerala's revenue deficit has remained sensitive to economic shocks and changing fiscal conditions, with noticeable improvement after the pandemic followed by a moderate rise in the most recent year.

Figure 3.2.10: Trend in Fiscal Deficit Ratio (% of GSDP) in Kerala (2005–2025)



The figure presents the trend in **Fiscal Deficit Ratio (Fiscal Deficit as a percentage of GSDP)** in Kerala during the period **2005–06 to 2024–25**. The ratio measures the overall fiscal deficit relative to the size of the state's economy and serves as an important indicator of fiscal sustainability.

Fiscal Deficit Ratio remained relatively stable during most of the study period, fluctuating between **2.46%** and **4.56%**. In the pre-GST years, the ratio generally remained around **3% to 3.6%**, indicating a fairly stable fiscal position. The ratio increased to **4.17%** in **2016–17** before moderating during the GST period.

A sharp increase was observed during the **COVID-19** period, when the Fiscal Deficit Ratio reached its highest value of **4.56%** in **2020–21**, reflecting increased government borrowing and expenditure to manage the economic crisis. The ratio subsequently declined to **2.46%** in **2022–23**, before rising again to **3.02%** in **2023–24** and **3.86%** in **2024–25**. Overall, the trend suggests that Kerala maintained a relatively stable fiscal deficit over most of the study period, with significant deviations occurring during periods of economic stress, particularly the pandemic.

3.3 Correlation Analysis

Correlation analysis was carried out using Pearson's correlation coefficient to examine the relationship among the selected fiscal indicators of Kerala during the period 2005–06 to 2024–25. The correlation coefficient ranges from -1 to $+1$, where values closer to $+1$ indicate a strong positive relationship, values close to -1 indicate a strong negative relationship, and values near 0 indicate little or no linear relationship.

Table 3.3.1: Correlation Matrix of Major Fiscal Indicators in Kerala (2005-06 to 2024-25)

	<i>OWN TAX REVENUE</i>	<i>OWN NON- TAX REVENUE</i>	<i>REVENUE RECEIPT</i>	<i>REVENUE EXPENDITURE</i>	<i>CAPITAL EXPENDITURE</i>	<i>TOTAL EXPENDITURE</i>	<i>REVENUE DEFICIT</i>	<i>FISCAL DEFICIT</i>	<i>GSDP AT CURRENT PRICES</i>
OWN TAX REVENUE	1								
OWN NON-TAX REVENUE	0.9742	1							
REVENUE RECEIPT	0.9867	0.9571	1						
REVENUE EXPENDITURE	0.9788	0.9437	0.9941	1					
CAPITAL EXPENDITURE	0.9571	0.8957	0.9751	0.9802	1				
TOTAL EXPENDITURE	0.9783	0.9403	0.9939	0.9997	0.9843	1			
REVENUE DEFICIT	0.7897	0.7355	0.8123	0.8705	0.8507	0.8700	1		
FISCAL DEFICIT	0.9227	0.8815	0.9265	0.9591	0.9458	0.9595	0.9579	1	
GSDP AT CURRENT PRICES	0.9961	0.9712	0.9857	0.9854	0.9605	0.9845	0.8300	0.9462	1

Table 3.3.1 shows that all selected fiscal indicators exhibit **positive correlations**, indicating that these variables generally moved in the same direction throughout the study period.

Revenue Receipt shows an **extremely strong positive correlation** with **Revenue Expenditure (0.9942)** and **Total Expenditure (0.9940)**, suggesting that increases in government revenue were accompanied by corresponding increases in expenditure. Similarly, **Revenue Expenditure** and **Total Expenditure** have the **highest correlation coefficient (0.9998)**, indicating an almost perfect positive relationship since revenue expenditure constitutes the major component of total government expenditure.

Own Tax Revenue is highly correlated with **Revenue Receipt (0.9868)**, **Revenue Expenditure (0.9788)**, **Total Expenditure (0.9783)** and **GSDP at Current Prices (0.9961)**. These strong relationships indicate that growth in the state's own tax collection has been closely associated with overall revenue mobilisation, expenditure expansion and economic growth during the study period.

Own Non-Tax Revenue also exhibits strong positive relationships with all major fiscal variables, with correlation coefficients ranging from **0.7355** to **0.9712**. Although these values are comparatively lower than those of Own Tax Revenue, they still indicate a substantial positive association with the state's fiscal performance.

Capital Expenditure demonstrates strong positive correlations with **Revenue Expenditure (0.9803)**, **Total Expenditure (0.9843)** and **Revenue Receipt (0.9751)**, suggesting that higher government revenues and overall expenditure supported increased investment in capital assets and infrastructure.

The **Revenue Deficit** shows positive correlations with all selected variables, with coefficients ranging from **0.7355** (Own Non-Tax Revenue) to **0.8706** (Revenue Expenditure and Total Expenditure). This indicates that as government expenditure and revenue expanded over time, the revenue deficit also tended to increase, reflecting persistent pressure on the revenue account despite growth in receipts.

Similarly, the **Fiscal Deficit** exhibits strong positive correlations with the remaining fiscal indicators, particularly **Revenue Deficit (0.9580)**, **Revenue Expenditure (0.9591)** and **Total Expenditure (0.9595)**. This suggests that rising expenditure levels were accompanied by higher borrowing requirements and fiscal deficits over the study period.

Overall, the correlation matrix indicates a **strong positive interrelationship among Kerala's fiscal indicators**, reflecting the simultaneous expansion of revenue, expenditure and economic activity over the study period. However, while correlation reveals the strength and direction of association, it **does not imply a causal relationship** between the variables.

3.4 Period-wise Analysis

To understand the changes in Kerala's fiscal performance under different economic and policy environments, the study period was classified into four phases: **Pre-GST (2005–06 to**

2016–17), GST (2017–18 to 2019–20), COVID-19 (2020–21 to 2021–22), and Recovery (2022–23 to 2024–25).

Table 3.4.1: Period-wise Average Fiscal Ratios in Kerala (2005–06 to 2024–25)

PERIOD	REVENUE RECEIPT RATIO	REVENUE EXPENDITURE RATIO	CAPITAL EXPENDITURE RATIO	TOTAL EXPENDITURE RATIO	REVENUE DEFICIT RATIO	FISCAL DEFICIT RATIO
PRE-GST	11.59	13.72	1.28	15.01	2.14	3.39
GST	11.57	13.71	1.30	15.00	2.14	3.39
COVID	12.63	15.90	1.92	17.83	3.27	4.30
RECOVERY	11.25	12.91	1.51	14.42	1.66	3.11

Table 3.4.1 indicates that the **Revenue Receipt Ratio** remained relatively stable during the Pre-GST (11.59%) and GST (11.57%) periods, increased to **12.63%** during the COVID-19 period, and declined to **11.25%** during the recovery period. The temporary rise during the pandemic reflects increased revenue mobilisation and financial support, while the subsequent decline suggests that the growth of GSDP outpaced the growth in revenue receipts during the recovery years.

The **Revenue Expenditure Ratio** showed a similar pattern, averaging **13.72%** in the Pre-GST period and **13.71%** during the GST period. It increased significantly to **15.90%** during COVID-19 due to higher government expenditure on healthcare, welfare measures, and economic support. During the recovery period, the ratio declined to **12.91%**, indicating improved expenditure management relative to economic growth.

The **Capital Expenditure Ratio** increased steadily across the four periods, rising from **1.28%** in the Pre-GST period to **1.30%** during GST, reaching **1.92%** during COVID-19, and remaining relatively high at **1.51%** during the recovery period. This suggests that despite fiscal pressures, the government continued to allocate resources towards capital formation and infrastructure development.

The **Total Expenditure Ratio** remained almost unchanged between the Pre-GST (**15.01%**) and GST (**15.00%**) periods but increased sharply to **17.83%** during the COVID-19 period. The ratio subsequently declined to **14.42%** during the recovery phase, reflecting the gradual withdrawal of pandemic-related expenditure and improved fiscal consolidation.

A similar pattern is observed for the deficit indicators. The **Revenue Deficit Ratio** remained stable at **2.14%** during both the Pre-GST and GST periods but increased substantially to **3.27%** during COVID-19 before declining to **1.66%** in the recovery period. Likewise, the **Fiscal Deficit Ratio** increased from **3.39%** during both the Pre-GST and GST periods to **4.30%** during COVID-19, before falling to **3.11%** during the recovery period.

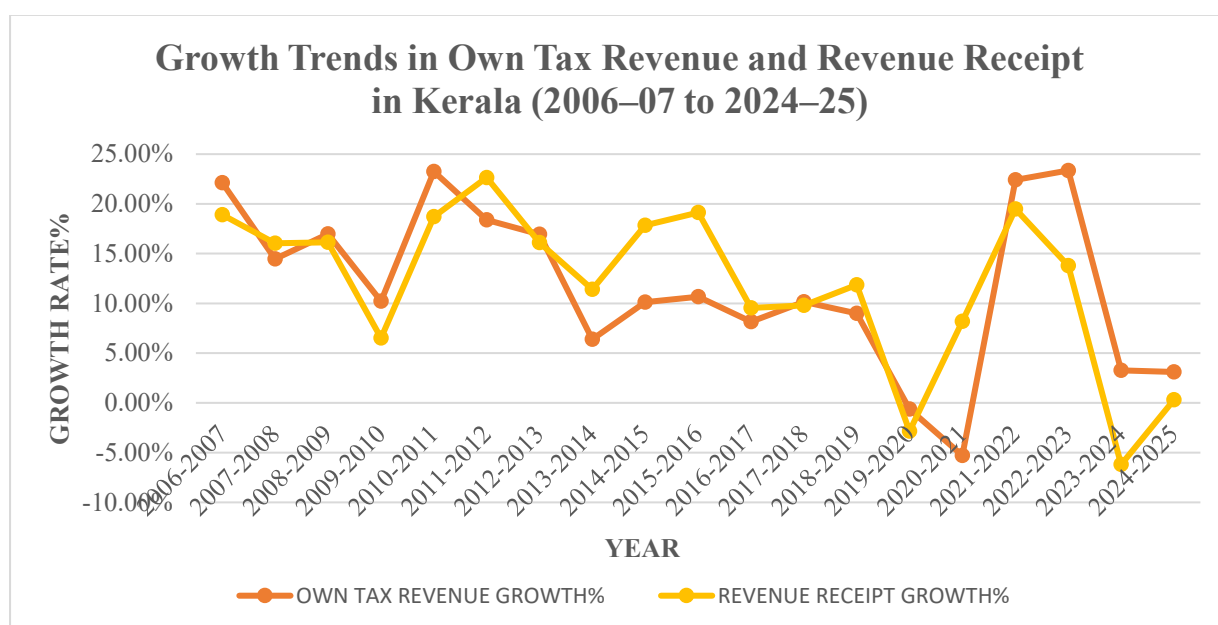
Overall, the period-wise analysis indicates that **the COVID-19 period experienced the greatest fiscal stress**, characterised by higher expenditure and larger fiscal deficits. In contrast, the recovery period shows evidence of **improved fiscal discipline**, with reductions in expenditure and deficit ratios while maintaining relatively higher capital expenditure

compared with the Pre-GST and GST periods. These findings suggest that Kerala gradually restored fiscal stability following the exceptional fiscal pressures created by the pandemic.

3.5 Growth Analysis

Growth analysis was carried out to examine the year-on-year changes in major fiscal indicators of Kerala during the period **2006–07 to 2024–25**. Annual growth rates were computed to identify periods of rapid expansion, slowdown and recovery in revenue generation, expenditure and fiscal balance. The analysis provides insights into the responsiveness of the state's fiscal indicators to policy changes, economic conditions and external shocks such as the COVID-19 pandemic.

Figure 3.5.1: Growth Trends in Own Tax Revenue and Revenue Receipt in Kerala (2006–07 to 2024–25)

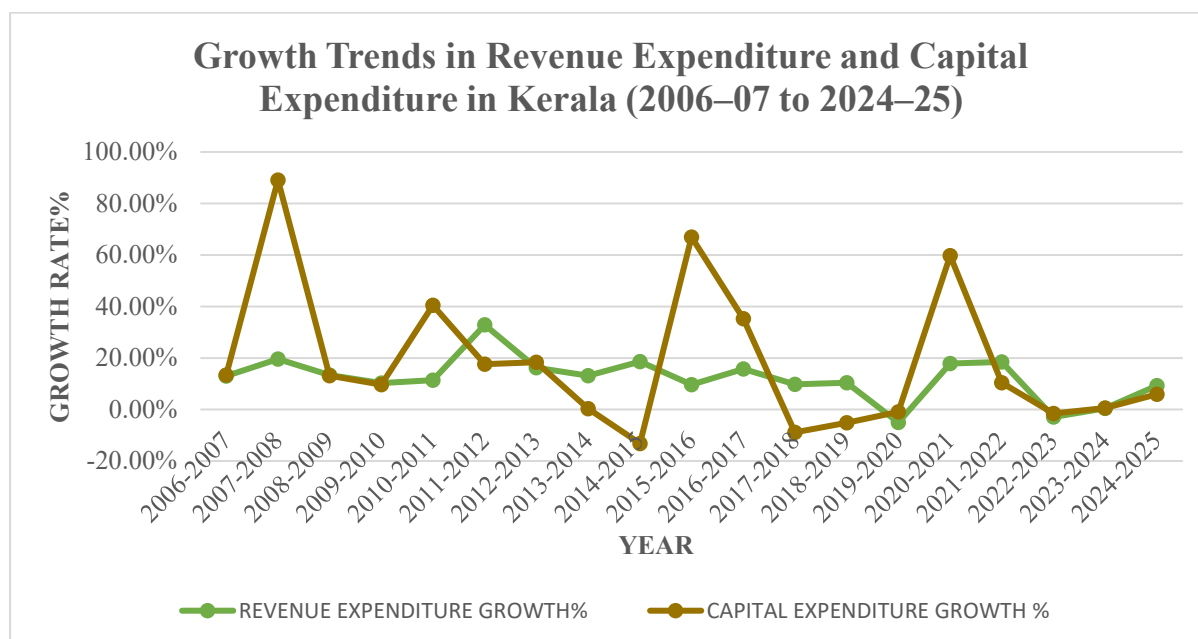


The figure compares the annual growth rates of **Own Tax Revenue (OTR)** and **Revenue Receipt (RR)** during the study period. Both indicators exhibited considerable fluctuations while maintaining an overall positive long-term growth trend.

Own Tax Revenue recorded its **highest annual growth of 23.36% in 2022–23**, reflecting a strong post-pandemic recovery in tax collections. In contrast, the **lowest growth (-5.29%)** occurred in **2020–21**, when economic activity was severely affected by the COVID-19 pandemic. Revenue Receipt followed a similar pattern, achieving its **highest growth of 22.65% in 2011–12**, while recording the **lowest growth (-6.21%) in 2023–24**.

The figure indicates that the growth patterns of Own Tax Revenue and Revenue Receipt generally moved together, although Own Tax Revenue experienced comparatively sharper fluctuations during certain years. The recovery observed after the pandemic highlights the resilience of Kerala's revenue mobilisation system and the gradual restoration of economic activity.

Figure 3.5.2: Growth Trends in Revenue Expenditure and Capital Expenditure in Kerala (2006–07 to 2024–25)



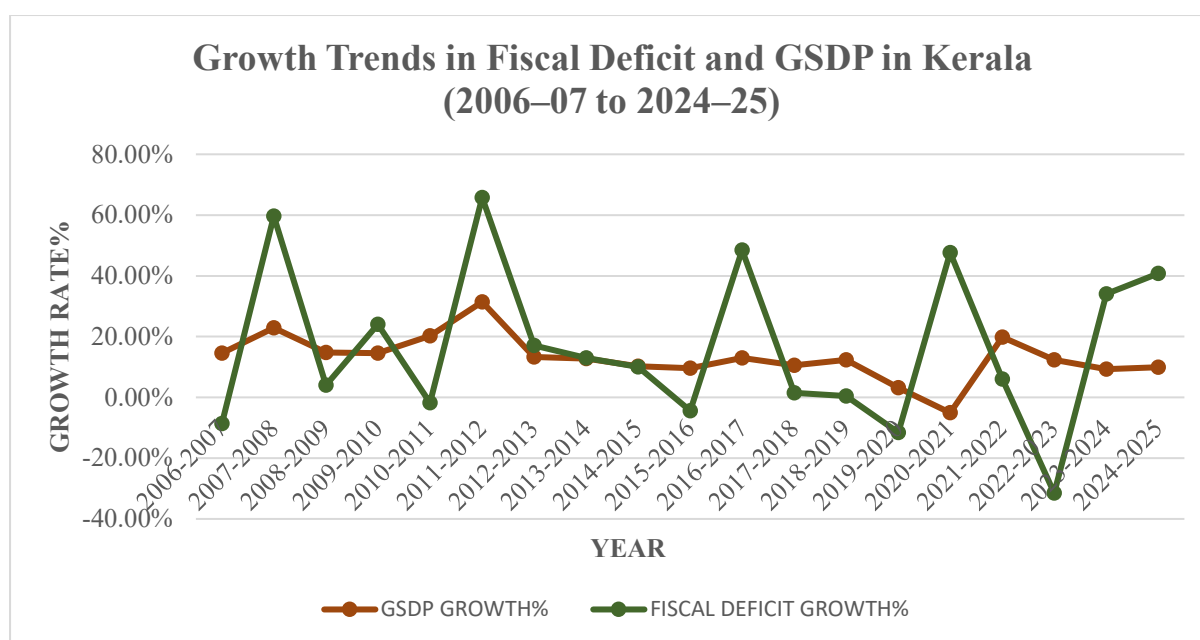
The figure illustrates the annual growth rates of **Revenue Expenditure** and **Capital Expenditure**. Revenue Expenditure displayed relatively moderate fluctuations throughout the study period compared with Capital Expenditure.

Revenue Expenditure recorded its **highest growth of 32.83% in 2011–12**, while the **lowest growth (-5.07%)** was observed in **2019–20**, reflecting expenditure moderation before the pandemic. During the COVID-19 period, Revenue Expenditure increased sharply, highlighting the government's response through higher spending on health services, welfare programmes and economic support measures.

Capital Expenditure exhibited substantially greater year-to-year volatility. The **highest annual growth (89.12%)** occurred in **2007–08**, indicating a significant expansion of investment expenditure during that year. Conversely, the **lowest growth (-13.21%)** was recorded in **2014–15**, reflecting a temporary reduction in capital investment. Another notable increase was observed during **2020–21**, demonstrating the state's continued emphasis on capital spending even during periods of fiscal stress.

Overall, the figure suggests that Revenue Expenditure remained comparatively stable, whereas Capital Expenditure responded more strongly to changes in fiscal priorities and investment decisions.

Figure 3.5.3: Growth Trends in Fiscal Deficit and GSDP in Kerala (2006–07 to 2024–25)



The figure presents the annual growth rates of **Fiscal Deficit** and **Gross State Domestic Product (GSDP)**. GSDP growth remained positive during most of the study period, although fluctuations were observed due to changes in economic conditions.

GSDP recorded its **highest annual growth of 31.43% in 2011–12**, while the **lowest growth (-5.07%)** occurred during **2020–21**, reflecting the severe economic disruption caused by the COVID-19 pandemic. Economic growth recovered in the subsequent years, indicating gradual restoration of economic activity.

Fiscal Deficit growth exhibited considerably higher volatility than GSDP growth. The **highest increase (65.77%)** occurred in **2011–12**, whereas the **largest decline (-31.50%)** was recorded in **2022–23**, reflecting a significant improvement in the state's fiscal position after the pandemic period. Despite these fluctuations, Fiscal Deficit growth generally remained more unstable than economic growth, indicating that fiscal balances were more sensitive to changes in government expenditure and revenue mobilisation.

Overall, the figure demonstrates that while Kerala's economy experienced relatively stable long-term growth, fiscal deficit growth was more volatile, particularly during periods of economic shocks and fiscal adjustment.

3.6 Fiscal Stress Score Analysis

Fiscal Stress Score is a composite indicator constructed using standardized (Z-score) values of selected fiscal variables to assess the overall fiscal condition of Kerala during the study period. The indicator combines revenue-side and expenditure-side fiscal ratios, enabling comparison of fiscal stress across years. Negative Fiscal Stress Score values indicate

relatively higher fiscal stress, whereas positive values indicate comparatively better fiscal conditions.

Table 3.6.1: Fiscal Stress Score and Ranking

YEAR	FISCAL STRESS SCORE	RANK
2005-2006	0.364026166	19
2006-2007	0.299438836	17
2007-2008	-0.069327294	8
2008-2009	-0.07234104	7
2009-2010	0.143285186	13
2010-2011	-0.044761304	9
2011-2012	0.241605437	15
2012-2013	0.130729291	12
2013-2014	0.312795404	18
2014-2015	0.443771014	20
2015-2016	-0.241018331	5
2016-2017	-0.489165455	2
2017-2018	-0.149944873	6
2018-2019	0.09650804	11
2019-2020	0.279946074	16
2020-2021	-0.631085405	1
2021-2022	-0.391219129	4
2022-2023	-0.455415734	3
2023-2024	0.005898968	10
2024-2025	0.226274152	14

Note: Fiscal Stress Scores were computed using standardized values (Z-scores) of Revenue Receipt Ratio, Revenue Expenditure Ratio, and Revenue Deficit Ratio. The mean and standard deviation used for standardization were calculated from the study period data.

Table 3.6.1 presents the Fiscal Stress Score and corresponding rank of Kerala for the period **2005–06 to 2024–25**. The Fiscal Stress Score was computed by combining standardized fiscal indicators, thereby providing a comprehensive measure of the state's fiscal condition over time. Lower (more negative) scores indicate periods of relatively higher fiscal stress, whereas higher (more positive) scores indicate comparatively better fiscal performance.

The analysis reveals that **2020–21** recorded the **highest level of fiscal stress**, with a Fiscal Stress Score of **–0.6311**, securing **Rank 1**. This period coincides with the COVID-19 pandemic, when the state experienced increased expenditure requirements, declining economic activity, and widening fiscal imbalances.

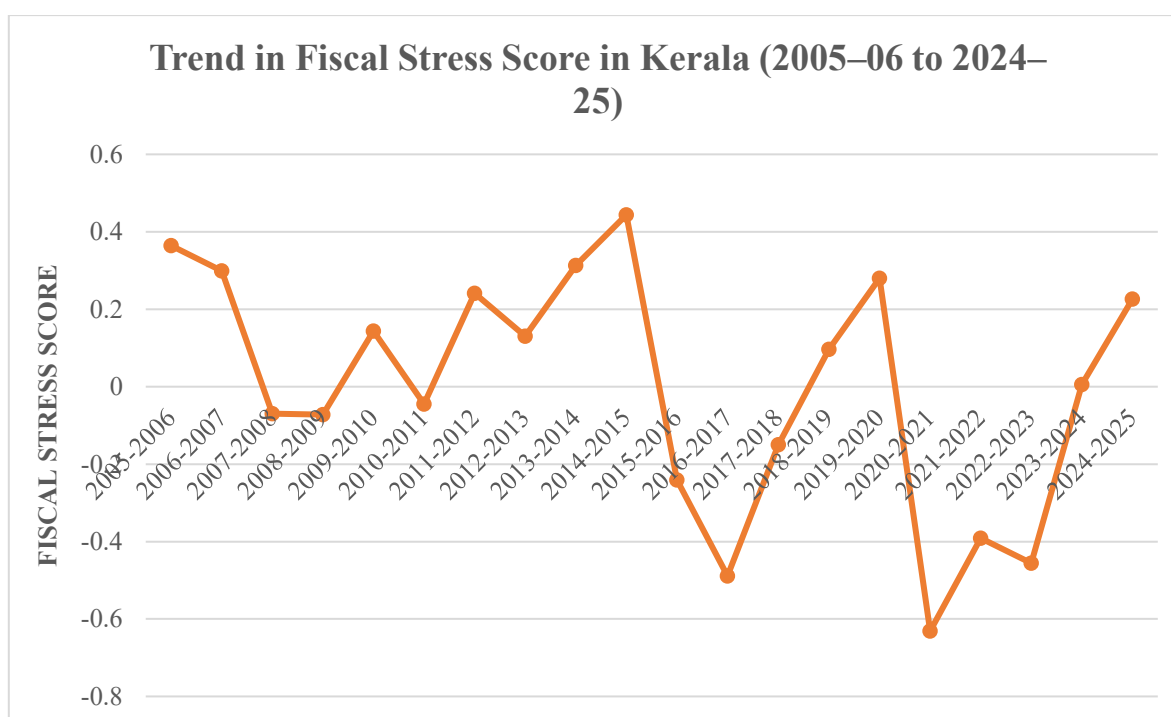
The second and third highest fiscal stress years were **2016–17** and **2022–23**, with Fiscal Stress Scores of **–0.4892** and **–0.4554**, respectively. These years also reflected considerable fiscal pressure arising from increased expenditure commitments and higher deficit levels. **2021–22**

and **2015–16** ranked fourth and fifth, respectively, indicating that fiscal stress remained elevated during the immediate post-pandemic period as well as during certain earlier years.

Conversely, **2014–15** exhibited the **lowest fiscal stress** during the study period, recording the highest Fiscal Stress Score of **0.4438** (Rank 20). This was followed by **2005–06**, **2013–14**, **2006–07**, and **2019–20**, which also showed relatively favourable fiscal conditions. These years were characterised by comparatively balanced fiscal indicators and lower deficit pressures.

Overall, the Fiscal Stress Scores indicate that Kerala's fiscal condition remained reasonably stable during much of the study period but deteriorated significantly during years associated with economic shocks and rising expenditure commitments. The ranking highlights the substantial fiscal pressure experienced during the COVID-19 period, followed by gradual improvement in subsequent years.

Figure 3.6.1: Trend of Fiscal Stress Score in Kerala (2005–06 to 2024–25)



The figure illustrates the annual trend in Kerala's Fiscal Stress Score during **2005–06 to 2024–25**. The Fiscal Stress Score fluctuated throughout the study period, reflecting changes in the state's fiscal position under varying economic and policy conditions.

The score remained largely positive during the initial years, indicating comparatively favourable fiscal conditions. However, notable declines were observed in **2015–16** and **2016–17**, signalling an increase in fiscal stress. The sharpest deterioration occurred during **2020–21**, when the Fiscal Stress Score reached its lowest value (**–0.6311**), reflecting the severe fiscal impact of the COVID-19 pandemic.

Although fiscal stress remained relatively high during **2021–22** and **2022–23**, the score improved gradually thereafter. By **2023–24**, the Fiscal Stress Score returned to a value close to

zero, and a further improvement was observed in **2024–25**, indicating a strengthening fiscal position during the recovery period.

Overall, the trend demonstrates that Kerala's fiscal stress has been influenced by major economic disruptions and expenditure pressures, while recent years suggest a gradual restoration of fiscal stability.

CHAPTER 4

FINDINGS AND CONCLUSIONS

4.1 Major Findings

The major findings of the study are summarized below:

1. **Own Tax Revenue** exhibited a consistent upward trend throughout the study period, increasing from ₹9,779.64 crore in 2005–06 to ₹76,642.20 crore in 2024–25. The highest annual growth of **23.36%** was recorded in **2022–23**, while the lowest growth (**-5.29%**) occurred during **2020–21**, reflecting the impact of the COVID-19 pandemic.
2. **Revenue Receipts** increased steadily over the study period, rising from ₹15,294.52 crore in 2005–06 to ₹132,724.65 crore in 2024–25. The highest annual growth (**22.65%**) was observed in **2011–12**, whereas the lowest growth (**-6.21%**) occurred in **2023–24**.
3. **Revenue Expenditure** showed continuous growth, increasing from ₹18,423.58 crore to ₹155,920.79 crore during the study period. Although expenditure continued to rise in absolute terms, the **Revenue Expenditure Ratio (% of GSDP)** generally declined, indicating improvement in expenditure management relative to the size of the state's economy.
4. **Capital Expenditure** increased substantially over time despite year-to-year fluctuations. The **Capital Expenditure Ratio (% of GSDP)** reached its highest value during **2020–21**, reflecting increased capital spending during the pandemic period.
5. **Revenue Deficit** and **Fiscal Deficit** both displayed an overall increasing trend, with noticeable spikes during the COVID-19 period. Revenue Deficit increased from **₹3,129.15 crore** in 2005–06 to **₹31,059.72 crore** in 2024–25, while Fiscal Deficit increased from **₹4,181.72 crore** to **₹48,248.14 crore** during the same period.
6. The **Revenue Receipt Ratio** remained relatively stable throughout the study period, fluctuating around **10–13% of GSDP**, indicating a fairly consistent revenue-generating capacity relative to the state's economic output.
7. The **Revenue Expenditure Ratio** generally declined from the pre-GST period to the recovery period despite temporary increases during COVID-19, suggesting improved expenditure efficiency relative to GSDP.
8. The **Revenue Deficit Ratio** and **Fiscal Deficit Ratio** exhibited considerable fluctuations across the study period, with the highest values recorded during the COVID-19 years, reflecting heightened fiscal pressure.
9. The **Capital Expenditure as a Percentage of Total Expenditure** increased gradually over the long run and remained above **10%** during the recent years, indicating greater emphasis on capital formation and infrastructure investment.
10. The **correlation analysis** revealed strong positive relationships among almost all fiscal variables. Revenue Receipts and Total Expenditure exhibited an almost perfect positive

correlation ($r = 0.9998$), while Revenue Receipts and Revenue Expenditure ($r = 0.9942$) and Own Tax Revenue and GSDP ($r = 0.9961$) also showed very high positive associations.

11. The **period-wise analysis** indicated that the **COVID period** recorded the highest average Revenue Expenditure Ratio (**15.90%**), Capital Expenditure Ratio (**1.92%**), Revenue Deficit Ratio (**3.27%**), and Fiscal Deficit Ratio (**4.30%**), demonstrating the significant fiscal impact of the pandemic.
12. The **Recovery period** showed improvement in fiscal conditions, with reductions in Revenue Deficit Ratio (**1.66%**) and Fiscal Deficit Ratio (**3.11%**) compared to the COVID period, indicating gradual restoration of fiscal stability.
13. Growth analysis revealed that most fiscal variables experienced temporary contractions during **2020–21**, followed by strong recovery in the subsequent years, particularly in Own Tax Revenue and Revenue Receipts.
14. The **Fiscal Stress Score Analysis** identified **2020–21** as the year with the highest fiscal stress, followed by **2016–17**, **2022–23**, **2021–22**, and **2015–16**. Conversely, **2014–15** recorded the lowest fiscal stress, indicating the most favourable fiscal position during the study period.
15. Overall, the study indicates that Kerala's fiscal performance improved considerably over the long term in terms of revenue mobilisation and economic growth. However, the state experienced substantial fiscal stress during periods of economic disruption, particularly the COVID-19 pandemic, highlighting the importance of prudent fiscal management and sustainable expenditure policies.

4.2 Conclusion

The present study examined the fiscal performance and fiscal stress of Kerala during the period **2005–06 to 2024–25** using descriptive statistics, trend analysis, correlation analysis, period-wise analysis, growth analysis, and a composite Fiscal Stress Score. The findings indicate that Kerala experienced substantial growth in revenue mobilisation, government expenditure, and Gross State Domestic Product (GSDP) over the study period.

Own Tax Revenue and Revenue Receipts showed a consistent upward trend, reflecting improvements in the state's revenue generation capacity. Although Revenue Expenditure and Capital Expenditure also increased considerably, their ratios to GSDP remained relatively stable or declined over time, suggesting improvements in fiscal management relative to the size of the economy.

The study also revealed that Revenue Deficit and Fiscal Deficit increased significantly during periods of economic disruption, particularly during the **COVID-19 pandemic**, when expenditure requirements increased sharply while revenue growth slowed. Period-wise analysis confirmed that the COVID period recorded the highest expenditure and deficit ratios, indicating severe fiscal pressure. However, the recovery period showed signs of improvement, with declining deficit ratios and strengthening revenue performance.

Correlation analysis demonstrated strong positive relationships among the major fiscal variables, indicating that revenue mobilisation, government expenditure, and economic growth have moved together over time. Growth analysis further highlighted the temporary contraction experienced during the pandemic, followed by a gradual recovery in subsequent years.

The Fiscal Stress Score successfully identified periods of relatively high and low fiscal stress. The analysis showed that **2020–21** experienced the highest fiscal stress, while **2014–15** represented the most favourable fiscal position during the study period. These findings demonstrate the usefulness of the composite Fiscal Stress Score as a tool for evaluating fiscal sustainability across different economic conditions.

Overall, the study concludes that Kerala has strengthened its fiscal capacity over the long term despite facing significant challenges arising from economic slowdowns and external shocks. Sustained efforts to enhance revenue mobilisation, maintain prudent expenditure management, and strengthen fiscal discipline will be essential for ensuring long-term fiscal sustainability and supporting inclusive economic development.

4.3 Suggestions

Based on the findings of the study, the following suggestions are proposed to strengthen Kerala's fiscal position and reduce fiscal stress:

1. **Strengthen revenue mobilisation:** The government should continue to improve tax administration, enhance compliance, and broaden the tax base to increase Own Tax Revenue without imposing excessive tax burdens.
2. **Promote non-tax revenue sources:** Greater emphasis may be placed on improving revenue from user charges, public sector enterprises, government services, and efficient management of public assets to diversify the state's revenue base.
3. **Maintain expenditure discipline:** Revenue expenditure should be monitored carefully to ensure that government spending remains efficient and sustainable while meeting developmental and welfare objectives.
4. **Increase productive capital expenditure:** A higher proportion of Total Expenditure may be allocated towards Capital Expenditure, particularly in infrastructure, healthcare, education, and other productive sectors that support long-term economic growth.
5. **Reduce fiscal deficits through prudent financial management:** Efforts should be made to contain Revenue Deficit and Fiscal Deficit by improving expenditure efficiency, strengthening revenue generation, and prioritising productive public investment.
6. **Develop fiscal buffers for economic shocks:** The experience of the COVID-19 pandemic highlights the need for adequate fiscal reserves and contingency planning to manage future economic crises without placing excessive pressure on public finances.

7. **Strengthen periodic fiscal monitoring:** Regular monitoring of key fiscal indicators through composite measures such as the Fiscal Stress Score can assist policymakers in identifying emerging fiscal pressures and taking timely corrective measures.
8. **Support sustainable economic growth:** Continued efforts to promote investment, employment generation, and industrial development can strengthen GSDP growth, thereby improving the state's fiscal capacity over the long term.

REFERENCES

Government of Kerala. (2008). *Economic Review 2008*. State Planning Board, Government of Kerala, Thiruvananthapuram.

Government of Kerala. (2010). *Economic Review 2010*. State Planning Board, Government of Kerala, Thiruvananthapuram.

Government of Kerala. (2011). *Economic Review 2011*. State Planning Board, Government of Kerala, Thiruvananthapuram.

Government of Kerala. (2016). *Economic Review 2016*. State Planning Board, Government of Kerala, Thiruvananthapuram.

Government of Kerala. (2022). *Economic Review 2022*. State Planning Board, Government of Kerala, Thiruvananthapuram.

Government of Kerala. (2025). *Economic Review 2025*. State Planning Board, Government of Kerala, Thiruvananthapuram.